UNITED STATES DEPARTMENT OF COMMERCE, Charles Sawyer, Secretary
NATIONAL BUREAU OF STANDARDS, E. U. Condon, Director

ATOMIC ENERGY LEVELS

As Derived From the Analyses of Optical Spectra

Volume I

The Spectra of Hydrogen, Deuterium, Tritium, Helium, Lithium, Beryllium, Boron, Carbon, Nitrogen, Oxygen, Fluorine, Neon, Sodium, Magnesium, Aluminum, Silicon, Phosphorus, Sulfur, Chlorine, Argon, Potassium, Calcium, Scandium, Titanium, and Vanadium

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HELIUM

He I

2 electrons $Z=2$

Ground state $1s^2\ {}^1S_0$

$1s^2\ {}^1S_0$ **198305±15** cm^{-1} I. P. 24.580 volts

Most of the terms are taken from Paschen-Götze with the term values subtracted from Paschen's limit as quoted by Robinson in 1937. Higher members of the $^1F°$ and $^3F°$ series are taken from Meggers and Dieke. The term $2p\ {}^3P°$ has been calculated from its combination with $2s\ {}^3S_1$, using the resolved triplet as observed by Meggers, the intervals being -0.078 cm^{-1} and -0.996 cm^{-1}. The components of $3p\ {}^3P°$ are based on Paschen's value of $3p\ {}^3P_2^2$ and the intervals observed by Gibbs and Kruger; -0.165 cm^{-1} and -0.192 cm^{-1}.

Some doubt exists regarding the correct classifications of lines attributed to doubly excited helium, such as those observed at 309.04 A and 320.38 A by Compton and Boyce, and at 320.392 A and 357.507 A by Kruger. Approximate theoretical computations of the energies of doubly excited levels have been made by a number of authors and are summarized by Wu. His classification of the line observed at 320.4 A as $2p\ {}^3P°-2p^2\ {}^3P$ has been adopted and used for the calculation of $2p^2\ {}^3P$.

Several references deal with intercombinations in He I, namely, those by Lyman, Hopfield, Paschen, Suga, and others. The term values based on the excellent long series have been adopted in the table, since it is believed that they are the most accurate.

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He I

Config.	Desig.	J	Level
$1s^2$	$1s^2\ ^1S$	0	0 ± 15
$1s\ 2s$	$2s\ ^3S$	1	159850. 318
$1s\ 2s$	$2s\ ^1S$	0	166271. 70
$1s\ 2p$	$2p\ ^3P^\circ$	2	*169081. 111*
		1	*169081. 189*
		0	*169082. 185*
$1s\ 2p$	$2p\ ^1P^\circ$	1	*171129. 148*
$1s\ 3s$	$3s\ ^3S$	1	183231. 08
$1s\ 3s$	$3s\ ^1S$	0	184859. 06
$1s\ 3p$	$3p\ ^3P^\circ$	2	*185558. 92*
		1	*185559. 085*
		0	*185559. 277*
$1s\ 3d$	$3d\ ^3D$	3, 2, 1	186095. 90
$1s\ 3d$	$3d\ ^1D$	2	186099. 22
$1s\ 3p$	$3p\ ^1P^\circ$	1	*186203. 62*
$1s\ 4s$	$4s\ ^3S$	1	190292. 46
$1s\ 4s$	$4s\ ^1S$	0	190934. 50
$1s\ 4p$	$4p\ ^3P^\circ$	2, 1, 0	*191211. 42*
$1s\ 4d$	$4d\ ^3D$	3, 2, 1	191438. 83
$1s\ 4d$	$4d\ ^1D$	2	191440. 71
$1s\ 4f$	$4f\ ^3F^\circ$	4, 3, 2	*191446. 61*
$1s\ 4f$	$4f\ ^1F^\circ$	3	*191447. 24*
$1s\ 4p$	$4p\ ^1P^\circ$	1	*191486. 95*
$1s\ 5s$	$5s\ ^3S$	1	193341. 33
$1s\ 5s$	$5s\ ^1S$	0	193657. 78
$1s\ 5p$	$5p\ ^3P^\circ$	2, 1, 0	*193795. 07*
$1s\ 5d$	$5d\ ^3D$	3, 2, 1	193911. 48
$1s\ 5d$	$5d\ ^1D$	2	193912. 54
$1s\ 5f$	$5f\ ^1F^\circ$	3	*193914. 31*
$1s\ 5f$	$5f\ ^3F^\circ$	4, 3, 2	*193915. 79*
$1s\ 5p$	$5p\ ^1P^\circ$	1	*193936. 75*
$1s\ 6s$	$6s\ ^3S$	1	194930. 46
$1s\ 6s$	$6s\ ^1S$	0	195109. 17
$1s\ 6p$	$6p\ ^3P^\circ$	2, 1, 0	*195187. 21*
$1s\ 6d$	$6d\ ^3D$	3, 2, 1	195254. 37
$1s\ 6d$	$6d\ ^1D$	2	195255. 02
$1s\ 6f$	$6f\ ^1F^\circ$	3	*195256. 7*
$1s\ 6f$	$6f\ ^3F^\circ$	4, 3, 2	*195256. 82*
$1s\ 6p$	$6p\ ^1P^\circ$	1	*195269. 17*
$1s\ 7s$	$7s\ ^3S$	1	195862. 63
$1s\ 7s$	$7s\ ^1S$	0	195973. 19
$1s\ 7p$	$7p\ ^3P^\circ$	2, 1, 0	*196021. 72*
$1s\ 7d$	$7d\ ^3D$	3, 2, 1	196064. 00
$1s\ 7d$	$7d\ ^1D$	2	196064. 31
$1s\ 7f$	$7f\ ^1F^\circ$	3	*196065. 4*
$1s\ 7f$	$7f\ ^3F^\circ$	4, 3, 2	*196065. 51*
$1s\ 7p$	$7p\ ^1P^\circ$	1	*196073. 41*
$1s\ 8s$	$8s\ ^3S$	1	196455. 79
$1s\ 8s$	$8s\ ^1S$	0	196529. 03
$1s\ 8p$	$8p\ ^3P^\circ$	2, 1, 0	*196561. 08*
$1s\ 8d$	$8d\ ^3D$	3, 2, 1	196589. 42
$1s\ 8d$	$8d\ ^1D$	2	196589. 73
$1s\ 8f$	$8f\ ^1F^\circ$	3	*196590. 3*
$1s\ 8f$	$8f\ ^3F^\circ$	4, 3, 2	*196590. 42*
$1s\ 8p$	$8p\ ^1P^\circ$	1	*196595. 56*
$1s\ 9s$	$9s\ ^3S$	1	196856. 37
$1s\ 9s$	$9s\ ^1S$	0	196907. 13
$1s\ 9p$	$9p\ ^3P^\circ$	2, 1, 0	*196929. 68*
$1s\ 9d$	$9d\ ^1D$	2	196949. 49
$1s\ 9d$	$9d\ ^3D$	3, 2, 1	196949. 63
$1s\ 9f$	$9f\ ^1F^\circ$	3	*196950. 3*
$1s\ 9f$	$9f\ ^3F^\circ$	4, 3, 2	*196950. 36*
$1s\ 9p$	$9p\ ^1P^\circ$	1	*196953. 95*
$1s\ 10s$	$10s\ ^3S$	1	197139. 76
$1s\ 10s$	$10s\ ^1S$	0	197176. 36
$1s\ 10p$	$10p\ ^3P^\circ$	2, 1, 0	*197192. 63*
$1s\ 10d$	$10d\ ^1D$	2	197207. 08
$1s\ 10d$	$10d\ ^3D$	3, 2, 1	197207. 30
$1s\ 10f$	$10f\ ^3F^\circ$	4, 3, 2	*197208. 0*
$1s\ 10p$	$10p\ ^1P^\circ$	1	*197210. 41*
$1s\ 11s$	$11s\ ^3S$	1	197347. 05
$1s\ 11p$	$11p\ ^3P^\circ$	2, 1, 0	*197386. 98*
$1s\ 11d$	$11d\ ^1D$	2	197397. 62
$1s\ 11d$	$11d\ ^3D$	3, 2, 1	197397. 75
$1s\ 11f$	$11f\ ^3F^\circ$	4, 3, 2	*197398. 6*
$1s\ 11p$	$11p\ ^1P^\circ$	1	*197400. 18*
$1s\ 12s$	$12s\ ^3S$	1	197503. 69
$1s\ 12s$	$12s\ ^1S$	0	197524. 26

He I—Continued

Config.	Desig.	J	Level
$1s\ 12p$	$12p\ ^3P^\circ$	2, 1, 0	*197534. 44*
$1s\ 12d$	$12d\ ^1D$	2	197542. 54
$1s\ 12d$	$12d\ ^3D$	3, 2, 1	197542. 67
$1s\ 12p$	$12p\ ^1P^\circ$	1	*197544. 56*
$1s\ 13s$	$13s\ ^3S$	1	197624. 98
$1s\ 13p$	$13p\ ^3P^\circ$	2, 1, 0	*197649. 07*
$1s\ 13s$	$13s\ ^1S$	0	197649. 78
$1s\ 13d$	$13d\ ^1D$	2	197655. 19
$1s\ 13d$	$13d\ ^3D$	3, 2, 1	197655. 47
$1s\ 13p$	$13p\ ^1P^\circ$	1	*197656. 95*
$1s\ 14s$	$14s\ ^3S$	1	197721. 13
$1s\ 14p$	$14p\ ^3P^\circ$	2, 1, 0	*197739. 90*
$1s\ 14d$	$14d\ ^1D$	2	197744. 918
$1s\ 14d$	$14d\ ^3D$	3, 2, 1	197744. 94
$1s\ 14p$	$14p\ ^1P^\circ$	1	*197746. 15*
$1s\ 15s$	$15s\ ^3S$	1	197796. 63
$1s\ 15p$	$15p\ ^3P^\circ$	2, 1, 0	*197813. 11*
$1s\ 15d$	$15d\ ^3D$	3, 2, 1	197817. 05
$1s\ 15p$	$15p\ ^1P^\circ$	1	*197818. 12*
$1s\ 16p$	$16p\ ^3P^\circ$	2, 1, 0	*197872. 95*

He I—Continued

Config.	Desig.	J	Level
$1s\ 16d$	$16d\ ^3D$	3, 2, 1	197876. 41
$1s\ 16p$	$16p\ ^1P^\circ$	1	*197877. 04*
$1s\ 17p$	$17p\ ^3P^\circ$	2, 1, 0	*197922. 51*
$1s\ 17d$	$17d\ ^3D$	3, 2, 1	197925. 33
$1s\ 17p$	$17p\ ^1P^\circ$	1	*197925. 87*
$1s\ 18p$	$18p\ ^3P^\circ$	2, 1, 0	*197964. 02*
$1s\ 18d$	$18d\ ^3D$	3, 2, 1	197966. 75
$1s\ 18p$	$18p\ ^1P^\circ$	1	*197966. 80*
$1s\ 19p$	$19p\ ^3P^\circ$	2, 1, 0	*197999. 12*
$1s\ 19d$	$19d\ ^3D$	3, 2, 1	198001. 43
$1s\ 19p$	$19p\ ^1P^\circ$	1	*198001. 44*
$1s\ 20p$	$20p\ ^3P^\circ$	2, 1, 0	*198029. 07*
$1s\ 20p$	$20p\ ^1P^\circ$	1	*198031. 02*
$1s\ 20d$	$20d\ ^3D$	3, 2, 1	198031. 41
$1s\ 21p$	$21p\ ^3P^\circ$	2, 1, 0	*198054. 83*
$1s\ 21d$	$21d\ ^3D$	3, 2, 1	198056. 50
$1s\ 22p$	$22p\ ^3P^\circ$	2, 1, 0	*198077. 15*
He II ($^2S_{1/2}$)	***Limit***		**198305**
$2p^2$	$2p^2\ ^3P$	2, 1, 0	481198

August 1946.

He II

(H sequence; 1 electron) $Z=2$

Ground state $1s\ ^2S_{1/2}$

$1s\ ^2S_{1/2}$ He^3 **438889.040** cm^{-1} I. P. He^3 54.400 volts

$1s\ ^2S_{1/2}$ He^4 **438908.670** cm^{-1} I. P. He^4 54.403 volts

The levels have been calculated by J. E. Mack, "using $R_{He^4}=109722.264$ and taking into account the fine structure as in hydrogen, but with $\Lambda=0.0402\pm0.009$, from the work of Skinner and Lamb on the 2*s*-level. The tentative experimental indication that Λ decreases with increasing n has been neglected. Assuming $R_{He^3}=109717.344$, the levels of He^3 may be calculated to a close approximation from those of He^4 by the equation

$$\text{Level}_{He^3\,II}-\text{Level}_{He^4\,II}=-(1-n^{-2})19.630\ cm^{-1}."$$

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Config.	Desig.	J	He^3 II Level	He^4 II Level	Interval
$1s$	$1s\ ^2S$	½	0.000	0.000	
$2p$	$2p\ ^2P°$	½	329164.390	329179.102	0.470
$2s$	$2s\ ^2S$	½	329164.860	329179.572	5.8434
$2p$	$2p\ ^2P°$	1½	329170.135	329184.945	
$3p$	$3p\ ^2P°$	½	390123.179	390140.622	0.14
$3s$	$3s\ ^2S$	½	390123.318	390140.761	1.7314
$3p$, $3d$	$3d\ ^2D$, $3p\ ^2P°$	1½	390124.910	390142.353	0.5771
$3d$	$3d\ ^2D$	2½	390125.487	390142.930	
$4p$	$4p\ ^2P°$	½	411458.517	411476.917	0.06
$4s$	$4s\ ^2S$	½	411458.576	411476.976	0.7304
$4p$, $4d$	$4d\ ^2D$, $4p\ ^2P°$	1½	411459.248	411477.648	0.2435
$4d$, $4f$	$4d\ ^2D$, $4f\ ^2F°$	2½	411459.491	411477.891	0.1217
$4f$	$4f\ ^2F°$	3½	411459.613	411478.013	
$5p$	$5p\ ^2P°$	½	421333.629	421352.472	0.03
$5s$	$5s\ ^2S$	½	421333.659	421352.502	0.3740
$5p$, $5d$	$5d\ ^2D$, $5p\ ^2P°$	1½	421334.003	421352.846	0.1247
$5d$, $5f$	$5d\ ^2D$, $5f\ ^2F°$	2½	421334.128	421352.971	0.0624
$5f$, $5g$	$5g\ ^2G$, $5f\ ^2F°$	3½	421334.190	421353.033	0.0374
$5g$	$5g\ ^2G$,	4½	421334.228	421353.071	
$6p$	$6p\ ^2P°$	½	426697.845	426716.928	0.02
$6s$	$6s\ ^2S$	½	426697.862	426716.945	0.2164
$6p$, $6d$	$6d\ ^2D$, $6p\ ^2P°$	1½	426698.062	426717.145	0.0721
$6d$, $6f$	$6d\ ^2D$, $6f\ ^2F°$	2½	426698.134	426717.217	0.0361
$6f$, $6g$	$6g\ ^2G$, $6f\ ^2F°$	3½	426698.170	426717.253	0.0216
$6g$, $6h$	$6g\ ^2G$, $6h\ ^2H°$	4½	426698.192	246717.275	0.0144
$6h$	$6h\ ^2H°$	5½	426698.206	426717.289	
$7s$, etc.	$7s\ ^2S$, etc.	½, etc.	----------	429951.508 to .741	
$8s$, etc.	$8s\ ^2S$, etc.	½, etc.	----------	432050.863 to1.023	
$9s$, etc.	$9s\ ^2S$, etc.	½, etc.	----------	433490.169 to .283	
$10s$, etc.	$10s\ ^2S$, etc.	½, etc.	----------	434519.693 to .777	
$11s$, etc.	$11s\ ^2S$, etc.	½, etc.	----------	435281.423 to .486	
$12s$, etc.	$12s\ ^2S$, etc.	½, etc.	----------	435860.778 to .828	
$13s$, etc.	$13s\ ^2S$, etc.	½, etc.	----------	436311.653 to .692	
$14s$, etc.	$14s\ ^2S$, etc.	½, etc.	----------	436669.407 to .439	
$15s$, etc.	$15s\ ^2S$, etc.	½, etc.	----------	436957.026 to 8.052	
	∞=*Limit*	----------	----------	**438908. 670**	

February 1949.